Name : Nidhi

Date : ______

Deep learning

Assignment - 2

Aim: To understand & perform forward propagation.

Dataset: Iris flower classification

Sepal length	width	Petal length	width	Class
5.1	3.5	1.4	0.2	Setosa
7.0	3.2	4.7	1.4	versicolor
6.3	3.3	6.0	2.5	Virginica
5.8	2.7	4.1	1.0	versicolor
5.0	3.4	1.5	0.2	setosa

for calculation, take first row:

$$x = [5.1, 3.5, 1.4, 0.2]$$

Input Neuron : 4.

x_1 = sepal length
x_2 = sepal width
x_3 = Petal length
x_4 = Petal width

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output neurons = 3

O_1 = setosa

O_2 = vericolor

O_3 = virginica.

<u>Network</u>:

Input (4) → Hidden (2) → output (3)

$Bias_1 = 0.10$, $Bias2 = 0.20$

Weights (H_1) = $w_1 = 0.10$, $w_3 = 0.20$, $w_5 = 0.15$, $w_7 = 0.30$

Weights (H_2) = $w_2 = 0.10$, $w_4 = 0.25$, $w_6 = 0.20$, $w_8 = 0.10$.

<u>Calculating H_1:</u>

$$ZH_1 = X_1 W_1 + X_2 W_2 + X_3 W_5 + X_4 W_7 + B_1$$

$$= (5.1)(0.10) + (3.5)(-0.20) + (1.4)(0.15) + (0.2)(3) + 0.10$$

$$= \boxed{ZH_1 = 0.18}$$

<u>Sigmoid of H_1:</u>

$$H_1 = \frac{1}{1 + e^{-0.18}} = 0.5449$$

Date :

- calculating H_2:

$$z_{H_2} = X_1 W_2 + X_2 W_4 + X_3 W_6 + X_4 W_8 + B_2$$
$$= (5.1)(-0.10) + (3.5)(0.25) + (0.2)(-10) + 0.20$$
$$\boxed{= z_{H_2} = 0.825}$$

- sigmoid for H_2:

$$H_2 = \frac{1}{1+e^{-0.825}} = 0.6952$$

$$H = [0.5449, 0.6952]$$

- output layer:

Secosa = $W9 = 0.30$, $W10 = 0.40$, Bias = 0.05
versicolor = $W11 = 0.20$, $W12 = 0.35$, Bias = 0.10
virginica = $W_{13} = 0.50$, $W_{14} = 0.10$, Bias = 0.05

for setosa. : $z = H_1 W_9 + H_2 W_{10} + 0.005$
$= -0.0646.$

for versicolor : $z = H_1 W_{11} + H_2 W_{12} + 0.10$
$= 0.2343$

for virginica : $z = H_1 W_{13} + H_2 W_{14} - 0.05$
$= 0.2920$

Date :

output : [-0.00646, 0.2343, 0.2920]

highest score = 0.2920.
predicted class = virginica
final class = virginica.